



Battery Replacement Instructions

GEHC FRU 5390238

1. PROCEDURE

This procedure is applicable for replacing the 12Vdc battery in the following panel:
GEHC P/N 5161287 Discovery MR750 Main Disconnect Panel which may also be referred to as the
DVMR Main Disconnect Panel.

Reminder:



RISK OF ELECTRICAL HAZARD

Ensure all power sources feeding this PDB panel have been turned off and securely locked-out and tagged-out using the GEHC and hospital Lock-Out/Tag-Out (LOTO) procedures.

Depending on local rules, use protective equipment and tools. (Gloves and Arc Flash mask)

Required Tools & Equipment

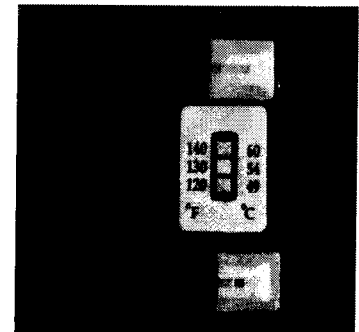
- Screwdrivers, various sizes, slotted & Phillips
- Digital Voltmeter

Familiarize yourself with this entire Installation Procedure before proceeding.

Save the FRU kit carton and packaging material to be used to ship the old battery to a recycling center.

2. BATTERY INSTALLATION

- Before installing the new battery, check the temperature sticker and measure the terminal voltage of the new battery. If the sticker shows more than 49°C or if the measured battery terminal voltage is less than 12Vdc, the battery is damaged or the storage conditions were not correct. The battery should be returned.
- If the temperature is not more than 49°C and the terminal voltage is 12Vdc or more, neatly write the Battery Installation Date on the battery label and continue.



FRU 5390238 Replacement Battery
Battery Factory Test Date: _____
Battery Installation Date: _____
1st Year Verification Date: _____
Additional Verification Date: _____
This battery must be replaced 2 years from the original installation date and every 2 years, thereafter.
SD-658-IM-RA Battery Install Date Label



- c. Open the enclosure door and locate the battery and battery mounting bracket.
- d. Turn OFF CB5, removing the load from the battery circuit.
- e. Disconnect the wires from the existing battery.

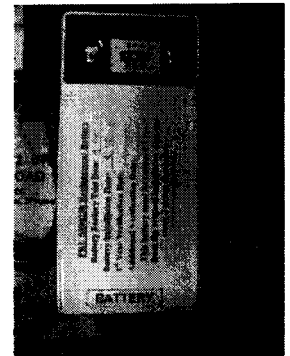
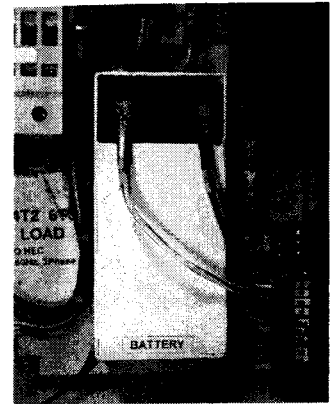
Dark Blue, wire #11, to the (+) RED battery terminal.

White w/ Blue stripe, wire #12, to the (-) BLACK battery terminal.

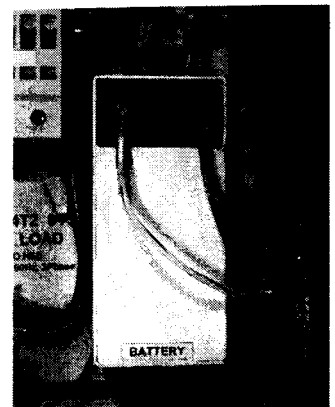
- f. Ensure these wires do not touch each other or any other conductor or grounded surface.
- g. Remove the (4) screws that hold the battery mounting bracket to the enclosure subpanel.
- h. Remove the bracket with the battery from the enclosure.
- i. Slide the old battery out of the bracket and set aside.

Note: There may be packing material installed between the battery and the inside of the bracket. This material, used to prevent the battery from shifting within the bracket, should be re-installed with the new battery.

- j. Carefully insert the new battery into the bracket, ensuring the terminals do not short to the bracket. Install the packing material to "shim" the battery in the bracket.
- k. Mount the bracket with the new battery into the enclosure.
- l. Apply the Installation Date label to the outside of the bracket.



- m. Remove and save the plastic cap(s) from the battery terminals.
- n. Re-connect the wires to the new battery.
- o. Dark Blue, wire #11, to the (+) RED battery terminal.
- p. White w/ Blue stripe, wire #12, to the (-) BLACK battery terminal.





3. PERFORM FUNCTIONAL TEST

- a. Ensure the Load equipment, the PDU and Heat Exchange Cabinet are turned off.
- b. Remove the Lock-out/Tag-out devices.
- c. Apply the 480 Vac source power to the panel. Turn the Main Circuit Breaker, CB1, ON. The front panel POWER ON light should illuminate.
- d. Turn CB5 to the ON position.
- e. Press the SYSTEM ON pushbutton. CR1 Should energize.
- f. Open CB4, which will remove the 120Vac from the charging circuit. The panel POWER ON light should extinguish but CR1 should remain energized.
- g. Press the SYSTEM OFF pushbutton. CR1 should de-energize.
- h. Place CB4 in the ON position.
- i. Monitor the battery terminal voltage for several minutes. Until the battery is fully charged, the terminal voltage will be increasing towards a final voltage of:
 - PULS P.S., full charge voltage: 13.45-13.55Vdc.
(The PULS P.S. green light should be on)
 - EDS 11027 Charger, full charge voltage:13.9-14.1Vdc
- j. Close and latch the panel door.
- k. Turn the load equipment on and place the system into operation.

4. DISPOSAL OF OLD BATTERY

- a. Install the plastic caps, that were removed from the new battery, on to the old battery terminals. If the cap(s) are damaged or missing, insulate the battery terminals with electricians tape.
- b. Dispose of the old battery responsibly. Re-use the FRU Kit packaging if shipping the battery to a recycling center.
- c. Use the supplied label to ensure the carton is marked with "NON SPILLABLE BATTERY".
- d. Refer to Appendix 3 for proper Battery Recycling

Appendix 1: Battery Data Sheet
Appendix 2: Battery Material Safety Data Sheet
Appendix 3: Battery Recycling
Appendix 4: Battery Declaration of EU Compliance

Rev	Description of Change	Changed By	Approved By	Date
A	Initial Release	JFH		



APENDIX 1 BATTERY DATA SHEET

Specific battery maintenance and recharge/discharge information can be found in attached battery manufacturer's data sheet.

PS-1270 12 Volt 7.0 AH
Rechargeable Sealed Lead Acid Battery

We've Got The Power.™

Features

- Absorbent Glass Mat (AGM) technology for superior performance
- Valve regulated, spill proof construction allows safe operation in any position
- Power/volume ratio yielding unrivaled energy density
- Rugged impact resistant ABS case and cover (UL94-HB)
- Approved for transport by air, D.O.T., I.A.T.A., F.A.A. and C.A.B. certified
- U.L. recognized under file number MH 20845

Terminals (mm)

F1 - Quick disconnect tabs, 0.187" x 0.032".
Mate with AMP, INC. FASTON "187" series

— OR —

F2 - Quick disconnect tabs, 0.250" x 0.032".
Mate with AMP, INC. FASTON "250" series

Performance Specifications

Nominal Voltage	12 volts (8 cells)
Nominal Capacity	
20-hr. (350mA to 10.50 volts)	7.00 AH
10-hr. (850mA to 10.50 volts)	6.50 AH
5-hr. (1.2A to 10.20 volts)	6.00 AH
1-hr. (4.5A to 9.00 volts)	4.50 AH
15-min. (14A to 8.00 volts)	3.50 AH
Approximate Weight	4.80 lbs. (2.18 kg)
Energy Density (20-hr. rate)	1.49 W-hr./in ³ (90.95 W-hr./l)
Specific Energy (20-hr. rate)	17.50 W-hr./lb (38.58 W-hr./kg)
Internal Resistance (approx.)	23 milliohms
Max Discharge Current (7 Min.)	21.0 amperes
Max Short-Duration Discharge Current (10 Sec.)	70.0 amperes
Shelf Life (% of nominal capacity at 68°F (20°C))	
1 Month	97%
3 Months	91%
6 Months	83%
Operating Temperature Range	
Charge	-4°F (-20°C) to 122°F (50°C)
Discharge	-40°F (-40°C) to 140°F (60°C)
Case	ABS Plastic
Power-Sonic Chargers	PSC-12800A, 12800A-C

Physical Dimensions: in (mm)

L: 5.95 (151) W: 2.56 (65) H: 3.70 (94) HT: 3.86 (98)

Tolerances are +/- 0.04 in (+/- 1mm) and +/- 0.06 in (+/- 2mm) for height dimensions. All data subject to change without notice.

To ensure safe and efficient operation always refer to the latest edition of our Technical Manual, as published on our website.
All data subject to change without notice.

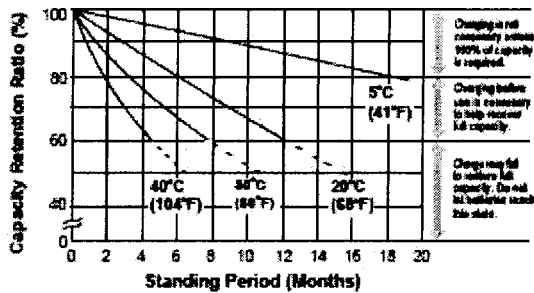
www.power-sonic.com



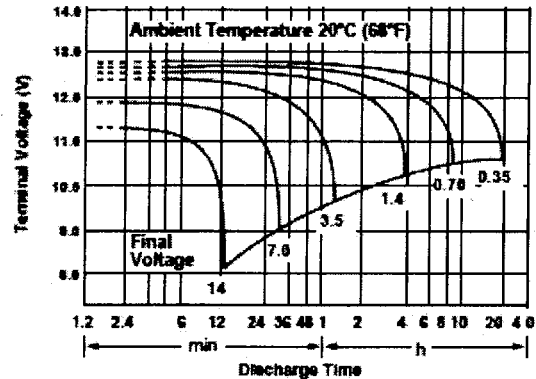
POWERSONIC

PS-1270 12 Volt 7.0 AH

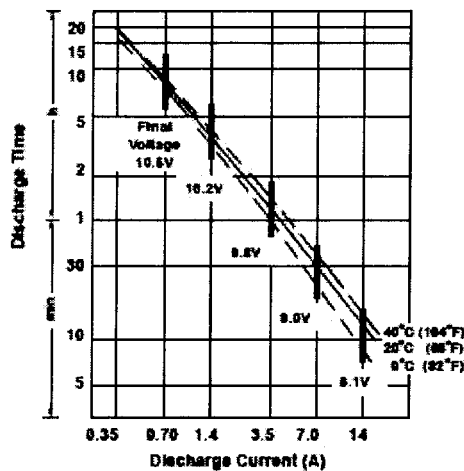
Shelf Life & Storage



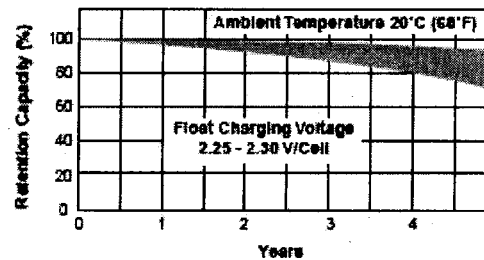
Discharge Characteristics



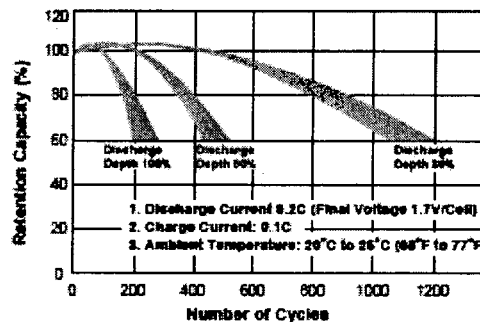
Discharge Time vs. Discharge Current



Life Characteristics in Stand-By Use



Life Characteristics in Cyclic Use



Charging

Cycle Applications: Limit initial current to 2.1A. Charge until battery voltage (under charge) reaches 14.4 to 14.7 volts at 68°F (20°C). Hold at 14.4 to 14.7 volts until current drops to under 70mA. Battery is fully charged under these conditions, and charger should be disconnected or switched to "float" voltage.

"Float" or "Stand-By" Service: Hold battery across constant voltage source of 13.5 to 13.8 volts continuously. When held at this voltage, the battery will seek its own current level and maintain itself in a fully charged condition.

Note: Due to the self-discharge characteristics of this type of battery, it is imperative that they be charged within 6 months of storage, otherwise permanent loss of capacity might occur as a result of sulfation.

Chargers

PowerSonic offers a wide range of chargers suitable for batteries up to 100AH. Please refer to the Charger Selection Guide in our specification sheets for "C-Series Switch Mode Chargers" and "Transformer Type A and F Series". Please contact our Technical department for advice if you have difficulty in locating suitable models.

Further Information

Please refer to our website www.power-sonic.com for a complete range of useful downloads, such as product catalogs, material safety data sheets (MSDS), ISO certification, etc..

Contact Information

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MSDS 04



APENDIX 2 BATTERY MANUFACTURER'S MATERIAL SAFETY DATA SHEET



Power-Sonic MSDS Page 1 of 6

MATERIAL SAFETY DATA SHEET

**PS, PSH, PSG, PG and Power Sport Series
Valve Regulated (VRLA) Batteries
Absorbed Electrolyte (AGM)**

Section 1 - Product Identification

Manufacturers Name Power-Sonic Corporation, 7550 Panasonic Way San Diego, CA 92154	Emergency Telephone Numbers: CHEMTREC (Domestic): (800) 424-9300 CHEMTREC (International): (703) 527-3887
	Telephone Number for Information Power-Sonic Corporation: (619) 661-2020
	Date Issued: June 10, 2008
<i>The information contained within is provided as a service to our customers and is for their information only. The information and recommendations set forth herein are made in good faith and are believed to be accurate at the date compiled. Power-Sonic Corporation makes no warranty expressed or implied.</i>	

Section 2 - Hazardous Ingredients/Identity Information

Components	CAS Number	Approx Wt. %	OSHA PEL ($\mu\text{g}/\text{m}^3$)	ACGIH TLV ($\mu\text{g}/\text{m}^3$)	NIOSH ($\mu\text{g}/\text{m}^3$)
Inorganic Lead/Lead Compounds	7439-92-1	65%-75%	50	150	10
Tin	7440-31-5	<0.5%	2000	2000	N/A
Calcium	7440-70-2	<0.1%	N/A	N/A	N/A
Electrolyte: Dilute sulfuric Acid	7664-93-9	14-20%	1000	1000	1000
Fiberglass Separator	-	5%	N/A	N/A	N/A
Case Material: Acrylonitrile Butadiene Styrene (ABS)	9003-56-9	5-10%	N/A	N/A	N/A

Inorganic lead and electrolyte (sulfuric acid) are the main components of every Valve Regulated Lead Acid battery supplied by Power-Sonic Corporation. Other ingredients may be present dependent upon the specific battery type. For additional information contact Power-Sonic Corporation Technical Department.

Section 3 - Physical/Chemical Characteristics

Components	Density	Melting Points	Solubility (H ₂ O)	Odor	Appearance
Lead	11.34	621 F°	None	None	Silver-Gray
Lead Sulfate	6.20	1950 F°	40mg/l(60 F°)	None	White Powder
Lead Dioxide	9.40	554 F°	None	None	Brown Powder
Sulfuric Acid	About 1.30	203-240 F°	100%	Sharp penetrating pungent	Clear Colorless Liquid
Fiberglass Separator	N/A	N/A	Slight	None	White Fibrous
Case Material: Acrylonitrile Butadiene Styrene (ABS)	N/A	N/A	None	None	Solid

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Section 4 – Flammability Data

Components	Flashpoint	Explosive Limit	Comments
Lead and Sulfuric Acid	None	None	None
Hydrogen		LEL = 4.1%	Sealed batteries can emit hydrogen if overcharged (float voltage > 2.40 VPC)
Fiberglass Separator	N/A	N/A	Toxic vapors may be released. In case of fire, wear self contained breathing apparatus
Acrylonitrile Butadiene Styrene (ABS)	None	N/A	Temp over 527°F (300°C) may release combustible gases. In case of fire, wear self contained breathing apparatus

Section 5 - Reactivity Data

Stability	Unstable		Conditions to Avoid
	Stable	X	Prolonged overcharge on high current, ignition sources. Sulfuric acid remains stable at all temperatures
Incompatibility (Materials to Avoid) Sulfuric acid: Contact with combustibles and organic materials may cause fire and explosion. Also reacts violently with strong reducing agents, metals, sulfur trioxide gas, strong oxidizers, and water. Contact with metals may produce toxic sulfur dioxide fumes and may release flammable hydrogen gas. Lead Compounds: Avoid contact with strong acids, bases, halides, halogenates, potassium nitrate, permanganate, peroxides, nascent hydrogen, and reducing agents.			
Hazardous Decomposition or Byproducts Sulfuric acid: Sulfur trioxide, carbon monoxide, sulfuric acid mist, sulfur dioxide, and hydrogen sulfide. Lead Compounds: High temperatures above the melting point are likely to produce toxic metal fume, vapor, or dust; contact with strong acid or base or presence of nascent hydrogen may generate highly toxic arsine gas. Hazardous Polymerization.			
Polymerization: Sulfuric acid will not polymerize			
Decomposition Products: Sulfuric Dioxide, Trioxide, Hydrogen Sulfide, Hydrogen.			
Conditions to Avoid: Prohibit smoking, sparks, etc. from battery charging area. Avoid mixing acid with other chemicals.			

Section 6 - Health Hazard Data

Routes of Entry Sulfuric acid: Harmful by all routes of entry Lead compounds: Hazardous Exposure can occur only when product is heated, oxidized, or otherwise processed or damaged to create dust, vapor or fume.
Inhalation Sulfuric Acid: Breathing sulfuric acid vapors and mists may cause severe respiratory problems. Lead Compounds: Dust or fumes may cause irritation of upper respiratory tract or lungs. Fiberglass Separator: Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F°/ use MSA Comfoil with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.
Skin Contact Sulfuric acid: Severe irritation, burns and ulceration. Lead Compounds: Not absorbed through the skin

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Ingestion Sulfuric acid: May cause severe irritation of the mouth, throat, esophagus, and stomach. Lead Compounds: May cause abdominal pain, nausea, vomiting, diarrhea, and severe cramping. Acute ingestion should be treated by a physician.
Eye Contact Sulfuric acid: Severe irritation, burns, cornea damage and possible blindness. Lead Compounds: May cause eye irritation.
Acute Health Hazards Sulfuric acid: Severe skin irritation, burns, damage to cornea may cause blindness, upper respiratory irritation. Lead Compounds: May cause abdominal pain, nausea, headaches, vomiting, loss of appetite, severe cramping, muscular aches and weakness, and difficulty sleeping. The toxic effects of lead are cumulative and slow to appear. It affects the kidneys, reproductive and central nervous systems. The symptoms of lead overexposure are listed above. Exposure to lead from a battery most often occurs during lead reclamation operations through the breathing or ingestion of lead dust or fumes.
Chronic Health Hazards Sulfuric acid: Possible scarring of the cornea, inflammation of the nose, throat and bronchial tubes, possible erosion of tooth enamel. Lead Compounds: May cause anemia, damage to kidneys and nervous system, and damage to reproductive system in both males and females.
Carcinogenicity Sulfuric acid: The National Toxicological Program (NTP) and The International Agency for Research on Cancer (IARC) have classified strong inorganic acid mist containing sulfuric acid as a Category 1 carcinogen, a substance that is carcinogenic to humans. The ACGIH has classified strong inorganic acid mist containing sulfuric acid as an A2 carcinogen (suspected human carcinogen). These classifications do not apply to liquid forms of sulfuric acid or sulfuric acid solutions contained within a battery. Inorganic acid mist (sulfuric acid mist) is not generated under normal use of this product. Misuse of the product, such as overcharging, may result in the generation of sulfuric acid mist. Lead Compounds: Human studies are inconclusive regarding lead exposure and an increased cancer risk. The EPA and the International Agency for Research on Cancer (IARC) have categorized lead and inorganic lead compounds as a B2 classification (probable/possible human carcinogen) based on sufficient animal evidence and inadequate human evidence.
Medical Conditions Generally Aggravated by Exposure Inorganic lead and its compounds can aggravate chronic forms of kidney, liver, and neurological diseases. Contact of battery electrolyte (acid) with the skin may aggravate skin diseases such as eczema and contact dermatitis. Overexposure to sulfuric acid mist may cause lung damage and aggravate pulmonary conditions.
Emergency and First Aid Procedures <u>Inhalation</u> Sulfuric acid: Remove to fresh air immediately. If breathing is difficult, give oxygen. Lead Compounds: Remove from exposure, gargle, wash nose and lips, consult physician. <u>Ingestion</u> Sulfuric acid: Do not induce vomiting, consult a physician immediately. Lead Compounds: Consult a physician immediately. <u>Eyes</u> Sulfuric acid: Flush immediately with water for 15 minutes, consult a physician. Lead Compounds: Flush immediately with water for 15 minutes, consult a physician. <u>Skin</u> Sulfuric acid: Flush with large amounts of water for at least 15 minutes, remove any contaminated clothing. If irritation develops seek medical attention. Lead Compounds: Wash with soap and water.

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Section 7 - Precautions for Safe Handling and Use

Steps to be Taken in Case Material is Released or Spilled There is no release of material unless the case is damaged or battery is misused/overcharged. If release occurs stop flow of material, contain/absorb all spills with dry sand, earth, or vermiculite. Do not use combustible materials. Neutralize spilled material with soda ash, sodium bicarbonate, lime, etc. Wear acid-resistant clothing, boots, gloves, and face shield. Dispose of as hazardous waste. Do not discharge acid to sewer.
Waste Disposal Method Spent Batteries - send to secondary lead smelter for recycling. Follow applicable federal, state and local regulations. Neutralize as in preceding step. Collect neutralized material in sealed container and handle as hazardous waste as applicable. A copy of this MSDS must be supplied to any scrap dealer or secondary lead smelter with the battery.
Precautions to be Taken in Handling and Storing Store batteries in a cool, dry, well ventilated area that are separated from incompatible materials and any activities which may generate flames, sparks, or heat. Keep all metallic articles that could contact the negative and positive terminals on a battery and create a short circuit condition.
Electrical Safety Due to the battery's low internal resistance and high power density, high levels of short circuit current can be developed across the battery terminals. Do not rest tools or cables on the battery. Use insulated tools only. Follow all installation instructions and diagrams when installing or maintaining battery systems.
Fiberglass Separator Fiberglass is an irritant to the upper respiratory tract, skin and eyes. For exposure up to 10F% use MSA Comfoil with type H filter. Above 10F use Ultra Twin with type H filter. This product is not considered carcinogenic by NTP or OSHA.

Section 8 - Control Measures

Respiratory Protection None required under normal conditions. If battery is overcharged and concentrations of sulfuric acid are known to exceed PEL use NIOSH or MSH approved respiratory protection.	
Engineering Controls Store and handle batteries in a well ventilated area. If mechanical ventilation is used, components must be acid resistant.	
Protective Gloves None needed under normal conditions. If battery case is damaged use rubber or plastic elbow length gauntlets.	Eye Protection None needed under normal conditions. If handling damaged or broken batteries use chemical splash goggles or face shield.
Other Protective Clothing or Equipment None needed under normal conditions. In case of damaged or broken battery use an acid resistant apron. Under severe exposure or emergency conditions wear acid resistant clothing.	
Work Hygienic Practices Handle batteries carefully to avoid damaging the case. Do not allow metallic articles to contact the battery terminals during handling. Avoid contact with the internal components of the battery.	

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Section 9 Regulatory Information

NFPA Hazard Rating for Sulfuric Acid Health=3 3 Flammability=0 2
Transportation Batteries: Non-Restricted Status North America Surface and Air Shipments Our nonspillable lead acid batteries are listed in the U.S. Department of Transportation's (DOT) hazardous materials regulations but are <u>excepted</u> from these regulations since they meet all of the following requirements found at 49 CFR 173.159(d) - NMFC # 60680 Class 65. • When offered for transport, the batteries are protected against short circuits and securely packaged as required by 49 CFR 173.159(d) (1); • The batteries and outer packaging are marked with the words NONSPILLABLE BATTERY as required by 49 CFR 173.159(d) (2); and • The batteries comply with the vibration and pressure differential tests found in 49 CFR 173.159(d) (3) and "crack test" found at 49 CFR 173.159(d) (4). International Our non-spillable lead acid batteries also are <u>excepted</u> from the international hazardous materials (also known as "dangerous goods") regulations since they comply with the following requirements: • The vibration and pressure differential tests found in Packing Instruction 806 and Special Provision A67 of the <u>International Air Transport Association (IATA) Dangerous Goods Regulations</u>. The vibration and pressure differential tests found in Packing Instruction 806 and Special Provision A67 of the International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air; and • The vibration, pressure differential, and "crack" tests found in Special Provision 238.1 and 238.2 of the <u>International Maritime Dangerous Goods (IMDG) Code</u>.
Regulatory Information RCRA: Spent lead acid batteries are not regulated as hazardous waste by the EPA when recycled, however state and international regulations may vary. CERCLA (superfund) and EPCRA: (a) Reportable Quantity (RQ) for spilled 100% sulfuric acid under CERCLA (superfund) and EPCRA (Emergency Planning Community Right to Know Act) is 1,000lbs. State and local reportable quantities for spilled sulfuric acid may vary. (b) Sulfuric acid is a listed "Extremely Hazardous Substance" under EPCRA with a Threshold Planning Quantity (TPQ) of 1,000lbs. (c) EPCRA Section 302 Notification is required if 1,000lbs. or more of sulfuric acid is present at one site. The quantity of sulfuric acid will vary by battery type. Contact Power-Sonic Corporation for additional information. (d) EPCRA Section 312 Tier 2 reporting is required for batteries if sulfuric acid is present in quantities of 500lbs. or more and/or lead is present in quantities of 10,000lbs. or more. (e) Supplier Notification: This product contains toxic chemicals which may be reportable under EPCRA Section 313 Toxic Chemical Release Inventory (Form R) requirements. If you are a manufacturing facility under SIC codes 20 through 39 the following information is provided to enable you to complete the required reports:

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Regulatory Information continued:

(f)

Toxic Chemical	CAS Number	Approximate % by weight
Lead	7439-92-1	60
Sulfuric Acid	7664-93-9 10-	30
Arsenic	7440-38-2	0.2

If you distribute this product to other manufacturers in SIC codes 20 through 39, this information must be provided with the first shipment in a calendar year. The Section 313 supplier notification requirement does not apply to batteries which are "consumer products". Not present in all battery types. Contact Power-Sonic Corporation for further information.

TSCA

Ingredients in Power-Sonic Corporation's batteries are listed in the TSCA Registry as follows:

Components	CAS Number	TSCA Status
Electrolyte Sulfuric Acid (H ₂ SO ₄)	7664-93-9	Listed
Inorganic Lead Compound: Lead (Pb)	7439-92-1	Listed
Lead Oxide (PbO)	1317-36-8	Listed
Lead Sulfate (PbSO ₄)	7446-14-2	Listed
Arsenic (As)	7440-38-2	Listed
Calcium (Ca)	7440-70-2	Listed
Tin (Sn)	7440-31-5	Listed

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APENDIX 3 BATTERY RECYCLING



Battery Recycling



Power-Sonic is dedicated to the protection of our environment and we would urge you to recycle your old batteries on every occasion.

We in the battery industry are proud of the fact that lead-acid batteries are the environmental success story of our time. More than 97% of all battery lead is recycled. Compared to 55% of aluminum soft drink and beer cans, 45% of newspapers, 26% of glass bottles and 26% of tires, lead acid batteries top the list of the most highly recycled consumer products.

The lead-acid battery gains its environmental edge from its closed loop cycle. The typical new lead-acid battery contains 60 to 80 percent recycled lead and plastic. When a spent battery is collected, it is sent to a permitted recycler, where under strict environmental regulations, the lead and plastic are reclaimed and sent to a new battery manufacturer. The recycling cycle goes on indefinitely. that means the lead and plastic in the lead-acid batteries that you use have been - and will continue to be - recycled many, many times. This makes lead acid battery disposal extremely successful from both environmental and cost perspectives.

To find out more about recycling, the applicable Federal and State laws and locate battery recycling locations in your area we would direct you to the following sites:

- <http://www.earth911.org> This is a very comprehensive site, and by entering your zip code the site will direct you to battery recycling depots in your area. Think and act green!
- <http://www.batteryCouncil.org> This site will provide you with more information on the recycling process and direct you to the pertinent Federal and State laws.





APENDIX 4 BATTERY DECLARATION OF EU COMPLIANCE



EU Battery Directive 2006/66/EC

Power-Sonic Corporation Sealed Lead Acid Batteries do not contain any of the substances contained in article 4 of the directive:

- (a) all batteries or accumulators, whether or not incorporated into appliances, that contain more than 0.0005% of mercury by weight; and
- (b) portable batteries or accumulators, including those incorporated into appliances, that contain more than 0.002% of cadmium by weight.

A copy of our Material Safety Data Sheet can be accessed from our website www.power-sonic.com. Page one of the document lists all the materials that are contained in our sealed lead acid batteries. It will be noted that the EU directive clearly states "no lead ban".

It is certified therefore that our sealed lead acid batteries are in full compliance with this directive.

May 15, 2008

